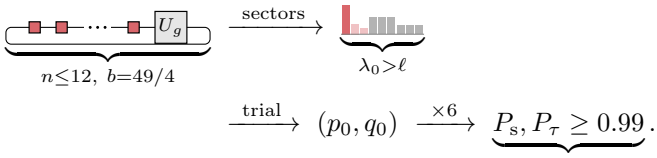
How the Initial Estimate Is Certified

(Dated: October 15, 2026)

The finite-scale input $P_{\mathrm{s}}(2304,784), P_{\tau}(4608,784) \geq 0.99$ of Ref. [1] is established without direct tensor contraction at that scale. Ref. [3] certifies a pair of bounds at length 36 and $\beta = 49/4$, which are then propagated to $(2304,784)$ through six scale doublings with decoupled defects. The underlying verification reduces to three closed tensor networks: twisted rings of at most twelve sites, symmetry-projected rings, and an integer matrix-product trial state of bond dimension 26 on 72 sites.

Appendix I: Architecture of the Certificate

Throughout, X_b is the spatial transfer operator of Ref. [1] at inverse temperature b, with the energy shift $\kappa = 700741/500000$ absorbed into H. Its eigenvalues $\lambda_i(b)$ are real, and $Z_n(b) = \sum_i \lambda_i^n$. The certificate proceeds in four steps, from twisted partition functions to the initial estimate:



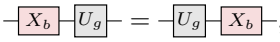
(I1)

- *Twisted partition functions* (Sec. J): certified values of $Z_n(b,g)$ for $4 \leq n \leq 12$, $b = 49/4$, and two twists g.
- *Sectors and filters* (Sec. K): bounds on every eigenvalue of X_b, sector by sector. A trial vector then forces one eigenvalue above $\ell = 0.999$.
- *Initial defects* (Sec. K): $1 - P_{\mathrm{s}}(36,b) \leq p_0$ and $1 - P_{\tau}(72,b) \leq q_0$.
- *Six doublings* (Sec. L): the coupled update reaches $(2304,784)$.

Every number is an exact rational or an integer; floating-point arithmetic is used nowhere in the proof.

Appendix J: Twisted boundary conditions

Twists and symmetries. A proper signed permutation g of the Cartesian axes acts on histories by U_g, which relabels events and multiplies by their signs. Rotating every matrix inside a site trace shows that U_g slides through a transfer vertex,



(J1)

so a single U_g may be placed on any virtual bond of the ring. On the physical side it conjugates one end of a single bond, the seam:

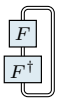
$$Z_n(b,g) = \operatorname{Tr} X_b^n U_g = \quad = \operatorname{Tr} e^{-bH_n(g)}. \tag{J2}$$



The certificate needs $g = 1$ and the π rotation $g = P$.

Purification. The physical side is computed as a squared Frobenius norm, which splits the imaginary-time circle at half its length,

$$Z_n(b,g) = \quad = \|F\|_F^2, \qquad F = e^{-bH/2}, \tag{J3}$$



so that only columns Fe_r are needed, one basis state e_r at a time.

Uniformization. Shift and scale H so that $R = 1 - (\frac{3}{2}n + H)/16$ is a contraction. Uniformization (Jensen's method) then writes F, up to the scalar $e^{-b(\kappa-3/2)n/2}$, as a Poisson mixture of powers of R:

$$F \propto e^{98(R-1)} = \sum_{j\geq 0} e^{-98} \frac{98^j}{j!} \quad , \tag{J4}$$



with j factors of R in the jth term. This is the discrete counterpart of the event expansion behind the thermal MPO: the events of W_β arrive as a Poisson process, and here each factor of R is one event.

The sum is cut at $j = 280$. The truncation error is below 2^{-84} in operator norm, because $\|R\| \leq 1$ [4]. The truncated sum is evaluated by Horner's rule on integer vectors scaled by $Q = 2^{55}$, taking the floor after every multiplication by R. Each multiplication adds at most $\sqrt{\dim} + 1$ to the Euclidean error, and the contraction carries earlier errors forward unamplified. Each column is therefore known to within $281(\sqrt{3^n} + 2)$ in units of $1/Q$.

Orbits. The trace in Eq. (J3) needs one column per basis word. Cyclic shifts, reversal and global spin flip permute the columns without changing their norms. For the twisted ring, a shift moves the seam, and a diagonal sign gauge moves it back. Only one representative per orbit is computed, weighted by the orbit size. The results are eighteen integers $W_{n,g}$ of about 30 digits. Turning them into rational enclosures gives the table of $Z_n(49/4, g)$, each value to within $\pm 3 \times 10^{-5}$. As an independent sanity check, not part of the proof, floating-point exact diagonalization reproduces all ten centers for $4 \leq n \leq 8$ to within 10^{-6}.

Appendix K: From partition functions to the transfer spectrum

Sector projectors. The four twists $g \in D_2 = \{1, P_x, P_y, P_z\}$ commute with X_b, so their character sums are projectors that can be inserted into the ring:

$$\Pi_\chi = \tfrac{1}{4}\textstyle\sum_{g\in D_2} \chi(g) U_g, \qquad \operatorname{Tr} X_b^n \Pi_\chi = \tag{K1}$$



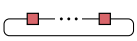
A cyclic rotation of the axes permutes the three nontrivial characters, so those three sectors carry identical eigenvalue lists, called N. Write J for the list of the invariant sector. Inverting the characters gives both lists from the two twists alone:

$$J_n = \tfrac{1}{4}\big[Z_n + 3Z_n(P)\big], \qquad N_n = \tfrac{1}{4}\big[Z_n - Z_n(P)\big], \tag{K2}$$

with $Z_n = J_n + 3N_n$. The coefficients sum to at most one in absolute value, so each sector moment inherits the 3×10^{-5} error. This holds for odd n too; odd moments carry sign information.

Polynomial filters (the term of Ref. [3]). For any polynomial $F \geq 0$ on $\mathbb{R}$, the closed network $\operatorname{Tr} F(X_b)$ is a finite combination of ring partition functions:

$$\sum_i F(\lambda_i) = \sum_j f_j \;\; \leq \; B. \tag{K3}$$



Every term on the left is nonnegative, so no single eigenvalue can have $F(\lambda) > B$. Two degree-12 filters of the form $x^4 Q(x)^2$ give

$$|\lambda| < v_J = 1.00139 \text{ in } J, \qquad |\lambda| < v_N = 0.872 \text{ in } N. \tag{K4}$$

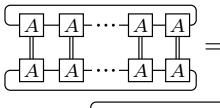
The filter needs $F \geq 0$ on the whole real line, not only on $[0, \infty)$, because X_b is self-adjoint but not positive. Reflection positivity is what makes the eigenvalues real [2].

Capped masses (also Ref. [3]'s term; a convexity bound). The twelfth moments are bounded by $m_J = 1.0657205$ and $m_N = 0.2783155$. Weights $t_i \leq h$ with total mass at most m have their rth powers maximized by filling bars to the cap,  :

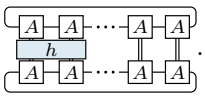
$$\sum_i t_i^r \leq \mathcal{C}(m, h, r) = qh^r + (m - qh)^r, \qquad q = \lfloor m/h \rfloor. \tag{K5}$$

The trial vector. One matrix-product vector on 72 sites, with real integer tensors A_s of bond dimension 26, has energy per site below $-\kappa$. Its norm and energy are two double-layer rings,

$$\langle\psi|\psi\rangle = \qquad = \operatorname{Tr} T^{72}, \tag{K6}$$



$$\langle\psi|H|\psi\rangle = 72$$



The virtual space carries four spin-$\frac{1}{2}$, three spin-$\frac{3}{2}$ and one spin-$\frac{5}{2}$ multiplets, $4{\cdot}2{+}3{\cdot}4{+}6 = 26$. The contraction splits into blocks of dimension at most 162, evaluated in exact integers. The outcome, $E_0(72) < -72\kappa$, means that one Boltzmann factor of $Z_{72}(b)$ exceeds one, so

$$Z_{72}(b) > 1 \quad \text{for every } b > 0. \tag{K7}$$

A dominant eigenvalue. Suppose every eigenvalue in J had $|\lambda| \leq \ell = 0.999$. The capped-mass bound applied to J and to the three copies of N would give $Z_{72}(49/4) \leq 0.931 < 1$, contradicting Eq. (K7). Hence some $\lambda_0 \in J$ has $|\lambda_0| > \ell$: the spectrum looks like , with one tall bar.

Removing λ_0 leaves the following bound on the kth moment of the rest:

$$R_A(k) = 3\,\mathcal{C}\big(m_N, v_N^{12}, \tfrac{k}{12}\big) + \big(m_J - \ell^{12}\big)^{k/12}. \tag{K8}$$

Initial defects. The two purities at the base scale now follow, each keeping only the dominant term:

$$P_{\mathrm{s}}(36, b) = \qquad \geq \Big(1 + \frac{R_A(36)}{\ell^{36}}\Big)^{-2} = 1 - p_0,$$



$$P_\tau(72, b) = \frac{Z_{72}(2b)}{Z_{72}(b)^2} \geq \big(v_J^{72} + R_A(72)\big)^{-2} = 1 - q_0. \tag{K9}$$

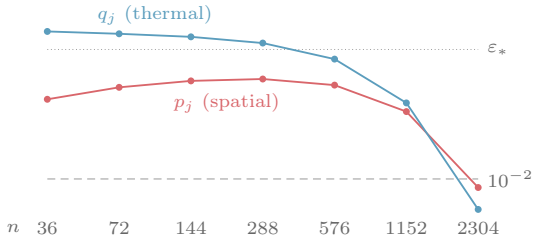
The first line uses $Z_{72} \geq \lambda_0^{72}$. The second uses $Z_{72}(2b) > 1$ from the trial vector. Numerically $p_0 = 0.0479$ and $q_0 = 0.1815$.

Appendix L: Six doublings

The thermal defect q_0 exceeds the diagonal threshold $\varepsilon_* \simeq 0.127$ of Ref. [2]. A single common ε would therefore not start the iteration. The coupled update keeps the two defects apart instead:

$$p_{j+1} = \big\lceil f\big(1 - (1 - p_j)^2(1 - q_j)\big)\big\rceil_7, \qquad q_{j+1} = \big\lceil f\big(1 - (1 - q_j)^2(1 - p_{j+1})\big)\big\rceil_7, \tag{L1}$$

These are moves 1–4 of the doubling diagram of Ref. [1], with each output rounded up to a multiple of 10^{-7}.



The figure plots the defects on a logarithmic scale against the length $n_j = 36 \cdot 2^j$; the inverse temperature doubles alongside.

- The thermal defect falls from the first step.
- The spatial defect first *rises*, to 0.071 at $j = 3$: at small n the coupling identity costs more than squaring gains.

- Both then collapse together and cross 10^{-2} at the last step,

$$\begin{aligned} 1 - P_{\mathrm{s}}(2304, 784) &\le 0.0084513, \\ 1 - P_{\tau}(4608, 784) &\le 0.0054986. \end{aligned} \tag{L2}$$

This recovers the initial estimate of Ref. [1] (Proposition 5.5 of Ref. [3]); re-evaluating the recurrence from the rational definitions of p_0 and q_0 confirms agreement digit for digit.

The weaker initial estimate and temperature lifting. Ref. [3] also certifies $P_{\mathrm{s}}(60, 105/4)$, $P_{\tau}(120, 105/4) > 7/8$ from a table at $b = 21/2$ for $n \le 10$, three twists, the $N = 120$ trial vector, and $\sum_m z_m^{5/2} \le (\sum_m z_m)^{5/2}$ for the Boltzmann factors $z_m = e^{-bE_m}$, giving $\Delta_L > \frac{4}{105} \log \frac{80}{79}$ for even $L \ge 60$. Checking the finite prefix intervals $[n_j, 2n_j]$ via $P_{\tau}(L, b_j) \ge (1-p_j)^2(1-q_j)$ and sharpening (v_J, v_N, ℓ) in Eq. (K9) to $(1.001351, 0.871256, 0.999998)$ already extends the $(36, 49/4)$ certificate to all even $L \ge 36$ with $\Delta_L > 0.01697$ (`models/spin1/experiments/`). Raising the inverse temperature directly on the $b_0 = 49/4$ sector moments before reducing to two purities gives a 19.1× larger bound than Ref. [1]: for even $12 \le k \le 60$ and $r \ge 1$, Eq. (K5) bounds $Z_k(b_0, P) \le J_k \le \mathcal{C}(m_J, v_J^{12}, k/12)$ and $Z_k(b_0) \le M_k := \mathcal{C}(m_J, v_J^{12}, k/12) + 3\,\mathcal{C}(m_N, v_N^{12}, k/12)$. Raising the positive Boltzmann factors of the self-adjoint twisted Hamiltonians to power r gives $Z_k(rb_0) \le M_k^r =: D$ and $Z_k(rb_0, P) \le \mathcal{C}(m_J, v_J^{12}, k/12)^r =: E$, so at $b = rb_0$ the spatial weights $u_i = |\lambda_i(b)|^k$ satisfy $\sum_i u_i \le D$, J-sector mass at most $\tau = (D + 3E)/4$, and $\sum_i u_i^{2h} = Z_{120}(b) > 1$ $(h = 60/k)$. Because each N eigenvalue has multiplicity three, any $t > D/3$ with $Dt^{2h-1} < 1$ forces $u_0 \in J$ with $u_0 > t$, leaving residual sth moment at most

$$R_s(t) = \max\left\{ 3\Big(\frac{D-t}{3}\Big)^s,\ (\tau - t)^s + 3\Big(\frac{D-\tau}{3}\Big)^s \right\}, \tag{L3}$$

upgraded whenever $(t')^{2h} + R_{2h}(t) < 1$. For the physical Boltzmann factors $z_m = e^{-b_0 E_m(120)}$, $Z_{120}(b_0) \le M_{120}$ and $z_0 > 1$ give $\sum_{m>0}(z_m/z_0)^r \le (M_{120} - 1)^r$, yielding

$$\begin{aligned} P_{\mathrm{s}}(60, rb_0) &\ge \Big(1 + \frac{R_h(t)}{t^h}\Big)^{-2}, \\ P_{\tau}(120, rb_0) &\ge \big(1 + (M_{120} - 1)^r\big)^{-2}. \end{aligned} \tag{L4}$$

Combined with the $(36, 49/4)$ prefix on $36 \le L \le 72$, Eq. (L4) certifies $\Delta_L > 0.07045 > 7/100$ at $(k, r) = (30, 2)$ and $\Delta_L > 0.07290$ at $(28, 17/8)$ for every even $L \ge 36$. With a single untwisted bound $Z_n(b_0) \le U$ and trial $Z_{2hn}(b_0) > 1$ $(h, r \ge 1)$, the same argument with $D = U^r$ and $t > D/2$ gives $P_{\mathrm{s}}(hn, rb_0) \ge [1 + ((D - t)/t)^h]^{-2}$ and $P_{\tau}(2hn, rb_0) \ge [1 + (U^{2h} - 1)^r]^{-2}$ whenever $t^{2h} + (D-t)^{2h} < 1$ (`models/spin2/scripts/one_trace_target.jl`).

Appendix M: Model-Dependent vs Universal Components

Each ingredient is a closed network, and only a few of them depend on the model.

- *Model independent:* the uniformization of Eq. (J4), the orbit reduction, the filters of Eq. (K3), the capped masses, and the doubling of Eq. (L1).
- *Model specific:*
  - the symmetry group that splits the spectrum into sectors, which for spin 1 makes N appear three times;
  - one good trial vector;
  - the sizes at which the filters isolate one eigenvalue.

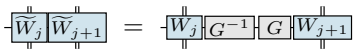
Smaller seed inputs for formal verification. Five studies in `models/lean_seeds` examine how to discharge the spin-1 seed inside the Lean 4 kernel (`lean/SeedKernel/`). First, even lengths alone suffice: $n \in \{4, 6, 8, 10, 12\}$ with the two real twists $g \in \{1, P\}$ at $b = 49/4 \to 49/2$ and trace error 10^{-3} certifies $\Delta_L > 0.039036$ for even $L \ge 60$, while $n \in \{6, 8, 10\}$ with twists $1, P, C$ (C the $2\pi/3$ axis permutation) at $b = 21/2$ and error 3×10^{-4} certifies $\Delta_L > 0.005966$ while cutting the orbit-trace dimension by 8.01×. Second, replacing Poisson uniformization by low-energy trial eigenstates  (as done unconditionally for the two-state Ising seed in Ref. [2]) and a half-chain comparison operator $H_n \succeq H_0 = A_{n/2} \otimes I + I \otimes A_{n/2} + 2cI$ (certified by exact rational 81×81 block $\mathrm{LDL}^{\mathsf{T}}$ factorization) checks $n = 12$ packed `Nat` matrix-vector products in 9.6–26 s. Unlike the Ising seed, however, spin 1 at $n = 12, b = 49/4$ has 26 $(g = 1)$ and 34 $(g = P)$ eigenstates below the tail cutoff, and the $S^z = 0$ comparison count (49 with a 32-bin tail vs. 8 physical levels) does not yet close without homotopy continuation, so a kernel-checked spin-1 seed remains open.

Spin-2 trial certificates and the conditional one-trace target. For spin 1, shift-add bond kernels and decomposing the $D = 26$ MPS transfer matrix into six SU(2) multiplicity blocks (largest dimension 56 vs. 162 in S^z) speed up the exact length-120 trial contraction by 360× (`models/spin1`). For spin 2 ($\boldsymbol{S}^2 = 6I$, $5^{12} \simeq 2.4 \times 10^8$ states, $\xi \approx 49.5$), the reflection-positive bond form and the bootstrap recurrence carry over unchanged, and SU(2)-invariant MPS trials certify $E_0(N)/N < -4.7595$ ($D = 24$, integer) and < -4.7612463 ($D = 295$, Arb; reduced by Schur's lemma and the π-rotation involution from $295^2 = 87{,}025$ to blocks of size at most 978) up to $N = 1024$ (`models/spin2`). By the untwisted one-trace bound above at $(n, b_0, h, r) = (256, 64, 2, 3)$ with shift $\kappa = 4.7612463$, a single thermal upper bound $Z_{256}(64) \le 1.08$ would certify $\Delta_L > 0.003019$ for even $L \ge 512$ (whereas

the $(128, 32)$ target at $\kappa = 4.76122$ is ruled out by the $N = 128$ trial lower bound $Z_{128}(32) > 1.1967$). Certifying $Z_{256}(64) \le 1.08$ remains open.

Certified thermal MPO compression. Two routes in `models/spin2` bound $Z_N(b)$ from above before MPO compression (`models/spin2/IDEAS.md`): shifting $K = H_N + NI \succeq 0$ ($\|K\| \le 5N$), a degree-d Taylor majorant $Q_d(K) = \sum_{j=0}^{d}(-\delta K)^j/j! + rI \succeq e^{-\delta K}$ ($\delta = b/(2m)$, $m = 2^p$) gives $Z_N(b) \le e^{(1-\kappa)bN}\,\mathrm{Tr}\,Q_d(K)^{2m}$, where the extensive norm $\delta\|K\| \le 5N\delta$ requires $p = 28$ squarings at $N = 256, b = 64$ for $d = 2$, or $p = 21$ for $d = 4$ (factored into a positive rational sum of squares of quadratics via the $\mathrm{LDL}^{\mathsf{T}}$ decomposition of the Laguerre moment matrix, $P_4(x) = (1 + \frac{x}{2} + \frac{x^2}{12})^2 + \frac{1}{12}(x + \frac{x^2}{2})^2 + \frac{1}{72}(x^2)^2$); alternatively, the Araki–Lieb–Thirring inequality for even and odd bond layers $H_N = A + B$ gives $Z_N(b) \le \mathrm{Tr}[(e^{-bA/m}e^{-bB/m})^m]$ with step error governed by the local bond norm rather than N. To certify MPO compressions along repeated squarings, `CertifiedGaps.CertifiedMPO` (`julia/src/CertifiedMPO/`) uses two tensor-network mechanisms:

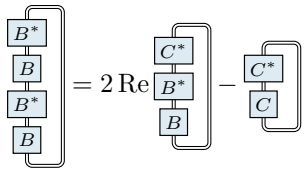
- *Rational matrix gauges.* Diagonal virtual-bond rescalings leave the local core-norm product $\prod_j \|W_j\|_{\mathrm{local}}$ at $N = 256$ a factor of 1.66×10^{75} above the analytic upper bound a_0 on $\|Q_2/5^{N/2}\|$, whereas invertible rational gauge matrices $\widetilde{W}_j = G_j W_j G_{j+1}^{-1}$ derived from the Schmidt decomposition cancel on every internal virtual bond,



(M1)

  bringing the Cholesky-certified product $\prod_j \|\widetilde{W}_j\|_{\mathrm{local}}$ within $1.000000701\,a_0$.

- *Three-layer trace correction and backward error propagation.* When a squared step $A = B^*B$ is compressed to C with $\|A - C\|_{\mathrm{F}} \le \eta$, the identity



(M2)

$$+ \|B^*B - C\|_{\mathrm{F}}^2$$

  replaces the four-layer trace $\mathrm{Tr}(A^2)$ and linear triangle error $(\|C\|_{\mathrm{F}} + \eta)^2$ by a three-layer signed overlap with quadratic residual η^2 (297× tighter single-step excess at $D = 32$). Across a multi-step flow, propagating approximate backward adjoints (`AdjointTrace.jl`) retains the signed linear errors across steps (1307× narrower $N = 16$ four-step enclosure), while backward Schatten bounds (`SchattenBridge.jl`) track $\|B_j\|_{p_j}$ with $p_j = 2^{k+1-j}$ to use operator-norm residuals at early steps and Hilbert–Schmidt residuals near the final trace. These certificates pass replay on 4- and 16-site flows and the first two squarings at $N = 256$ ($b = 2^{-20}$), leaving 26 squarings to $b = 64$ open.

[1] Companion note, *A diagrammatic proof of the Haldane gap via spacetime tensor networks*, `notes/thermal-mpo-gap-note.tex` (2026).

[2] Companion note, *Scope of the spacetime bootstrap*, `notes/thermal-mpo-gap-scope.tex` (2026).

[3] OpenAI, *The periodic spin-one Haldane gap*, manuscript dated September 24, 2026, Secs. 3–5 and Appendices A–C.

[4] B. L. Fox and P. W. Glynn, Computing Poisson probabilities, Commun. ACM **31**, 440 (1988).